

Transformations of $f(x) = |x|$, Part II - Practice

Algebra II | Unit 2 | Lesson 3 Practice | TEK A2.6.C

Objective: I can find the vertex of $f(x) = a|x-c|+d$ and write an equation given transformation information.

Section 1 — Recall

1. Give the vertex of $f(x) = |x - 6|$.

2. Give the vertex of $f(x) = |x| + 7$.

Section 2 — Apply

3. Give the vertex of $f(x) = 3|x + 2| - 5$.

4. Write a function with vertex $(4, -1)$, a vertical stretch by 2, and no reflection.

Section 3 — Challenge

5. Two functions have vertices $(2, 3)$ and $(-2, 3)$ - mirror images across the y-axis. Both have $a = 1$ and no reflection. Write both equations.

$f(x) =$ _____ $g(x) =$ _____

6. A graph opens downward with vertex $(-3, 6)$. What can you conclude about a ? Can you determine its exact value from this alone?

Section 4 — Explain It

7. Explain why the vertex of $f(x) = a|x - c| + d$ is always located exactly at (c, d) , no matter what a is.
